RangeRectQuant: Compressing Scientific Fields with Accurate Regional Sums

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Abstract

An absolute error budget for every future rectangle can force exact integer reconstruction when certified through independent pixel tolerances. RangeRectQuant coordinates errors within the valid pixel range while certifying all rectangles at once. It bounds the cumulative error from the top-left corner to each pixel. Four such prefix errors determine any rectangular error. The key step exactly characterizes which bounded prefix errors still permit completing a row with valid pixels and designated samples unchanged. A backward pass finds these states. A forward pass selects coarse multiples whenever they permit completion, retaining fine precision at constrained positions. This construction guarantees the prescribed error budget for every axis-aligned rectangular sum, with linear work in the number of pixels and row-length additional workspace. We characterize the fine choices and connect them to lossless representation length. Experiments on geographically held-out built-surface fields and water-occurrence maps show complete-file savings across regional-error budgets and field sizes. One decoded field supports accurate later measurements without the source or query-specific correction.

Introduction

Many bounded integer fields are stored before the regions to be measured are known. Later analyses may total values over neighborhoods, tiles, or other grid-aligned rectangles. A built-surface map, for example, records building area per pixel. The decoded values must remain between zero and the pixel's area, even after the source is discarded. We seek one absolute error budget for every rectangular sum, with designated samples reconstructed exactly.

Certifying these measurements through independent pixel tolerances can eliminate lossy integer reconstruction. Let A count samples allowed to change, U be the largest valid integer, $\delta \geq 0$ a finite uniform pixel tolerance, and ε the regional budget.

Proposition 1 (Independent pixel bounds)**.** *For integer source and output values in* $[0, U]$*, the sharp worst-case rectangular error under independent pixel tolerance* δ *is* $A \min(\lfloor \delta \rfloor, U)$*. If* $U \geq 1$ *and* $\varepsilon < A$*, these bounds certify the budget only with exact reconstruction.*

Proof. Set $d = \min(\lfloor \delta \rfloor, U)$. Each change has magnitude at most d. A zero source and output d at every unprotected sample attain Ad over the whole field. For $\varepsilon < A$, integer d must be zero. □

Coordinated changes can cancel these errors. A prefix sum adds values from the top-left corner to a pixel; four such sums determine any rectangular total [1, 2]. Bounding their errors gives every region an area-independent budget. The difficulty is realizing bounded prefix errors through valid pixels. Zero pixels cannot decrease, and fully built pixels cannot increase. Independently rounded prefixes can violate this range. Clipping changes the regional errors. A choice valid at the current pixel can also leave no legal way to finish the row.

RangeRectQuant exactly characterizes the bounded prefix-error states from which a row can still be completed with valid pixels (Figure 1). A backward pass represents this set by interval endpoints. A forward pass selects multiples of an integer step, called *coarse pixels*, whenever they permit completion. Other integers retain *fine precision*. This exact remaining-row feasibility yields a linear-time construction with one deterministic budget for every future rectangle. We connect its fine choices to lossless coding length and test complete files on built-surface and water-occurrence maps. Regional accuracy, the value range, and designated exact samples hold together in the delivered field.

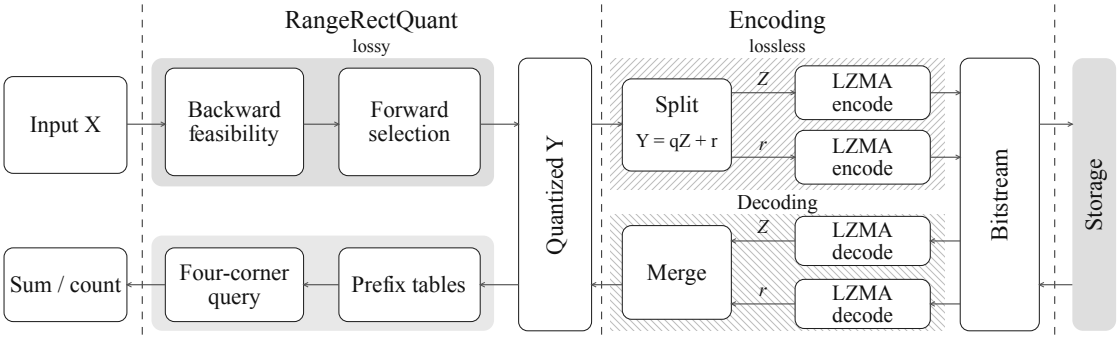


Figure 1: Overview of RangeRectQuant. Backward feasibility and forward selection turn input X into Y. For positive integer step q, quotient $Z = \lfloor Y/q \rfloor$ and remainder $r = Y - qZ$ are coded losslessly with the Lempel–Ziv–Markov chain algorithm (LZMA). Decoding restores Y; prefix tables of values and the exact missing mask serve regional sums and valid counts.

Related Work

Scientific compression balances precision and storage through prediction, quantization, and encoding [3]. SZ3 bounds individual pixel errors [4]; quality-driven interpolation improves rates under selected reconstruction metrics [5, 6]. Time-series prediction [7], quantization-error encoding [8], and integer representations for astronomical images [9] further reduce storage. Proposition 1 identifies the cost of using independent tolerances to certify all regional sums. RangeRectQuant coordinates pixel changes to meet that regional requirement directly.

Derived measurements, such as averages and totals, are called quantities of interest [10]. MGARD bounds field norms and errors in derived measurements [11, 12]. Quantity-preserving compression derives pixel tolerances for specified calculations [13]; QPET adds bound tuning, validation, and correction [14]. Quantizers also protect

selected ranges and regions [15] or topological features [16]. TOPIQ predicts bias and uncertainty for later queries [17]. Our target is a deterministic budget for every later rectangle, realized by one integer field that also obeys its physical range.

Coordinated rounding connects regional accuracy to reconstructed values. Wavelet summaries minimize mean-squared range-query error [18]. Matrix rounding uses floor or ceiling values to optimize regional errors [1]; its polynomial constructions cover two families of nested or disjoint regions. Lyu and Wang's two-dimensional sigma–delta quantizer bounds prefix errors [2]. Its stable greedy rule allows values beyond the input range, and clipping changes the errors supplying the guarantee. RangeRectQuant characterizes realizability with the range and exact designated values inside each transition. Its backward intervals retain precisely the states admitting a legal remaining row, enabling coarse forward choices without losing this completion guarantee.

Constructing a Field with Accurate Regional Sums

We construct a legal field, then encode its coarse values and remainders losslessly.

Problem definition

Let X be a field of H rows and W columns, with integer samples in $[0, U]$, where U is a nonnegative integer. Indices start at one. A mask marks missing samples, represented by zero in both X and the reconstruction Y. The protected set P includes these samples and any designated exact values. For a finite nonnegative budget ε, we require

$$Y_{ij} \in \{0, \ldots, U\}, \qquad Y_{ij} = X_{ij} \ ((i,j) \in P), \qquad \max_R \left| \sum_{(i,j)\in R} (Y_{ij} - X_{ij}) \right| \le \varepsilon, \quad (1)$$

where R ranges over all nonempty axis-aligned rectangles. Exact transmission of the mask also preserves each region's valid-sample count. A mean over $k > 0$ valid samples then has error at most ε/k.

Define the prefix error E_{ij} as the sum of $Y - X$ over rows 1 through i and columns 1 through j. Set $E_{0j} = E_{i0} = 0$, and let $B = \lfloor \varepsilon/4 \rfloor$. For $R = [a, b] \times [c, d]$,

$$\sum_R (Y - X) = E_{bd} - E_{a-1,d} - E_{b,c-1} + E_{a-1,c-1}. \quad (2)$$

Thus $|E_{ij}| \le B$ implies the regional budget in (1), independently of region size. Our construction keeps this bound while selecting values of Y directly.

The factor four is sharp even for binary pixels. For integer $B \ge 1$, repeat error row $(1, -1, -1)$ B times and its negative $2B$ times. Set source and output to one at negative and positive errors, respectively, and zero elsewhere. Every prefix error has magnitude at most B; the last $2B$ rows and last two columns have error $4B$.

A row transition that includes the value range

For row i, set $x_j = X_{ij}$ and $u_j = E_{i-1,j}$, with $u_0 = 0$. We construct the row from left to right. The state before the current pixel is $a = E_{i,j-1}$; its new state is $b = E_{ij}$. Define $\delta_j = u_j - u_{j-1}$ and $t_j = x_j - \delta_j$. The reconstructed pixel is $y_j = t_j + b - a$. The prefix bound requires $b \in [-B, B]$, the range requires $b \in [a - t_j, a + U - t_j]$, and a protected pixel forces $b = a + \delta_j$.

We favor values divisible by a positive integer step q, called *coarse pixels*. A *fine pixel* retains a nonzero remainder modulo q. This transition expresses the coarse condition as

$$b \equiv a - t_j \pmod{q}. \tag{3}$$

Copying the source row gives a feasible path: $a = u_{j-1}$ and $b = u_j$ produce $y_j = x_j$.

Exactly realizable prefix-error states

Let V_j contain the integer states after column j that allow the remaining row to be completed. Write $\mathbb{Z}$ for the integers and $P_i = \{j : (i, j) \in P\}$. Starting with $V_W = [-B, B] \cap \mathbb{Z}$, each interval $V_j = [L_j, H_j] \cap \mathbb{Z}$ has predecessor

$$V_{j-1} = \begin{cases} [-B, B] \cap [L_j + t_j - U, H_j + t_j] \cap \mathbb{Z}, & j \notin P_i, \\ [-B, B] \cap [L_j - \delta_j, H_j - \delta_j] \cap \mathbb{Z}, & j \in P_i. \end{cases} \tag{4}$$

A backward pass computes these endpoints. During the forward pass, the unprotected pixel's allowed states are

$$I_j = V_j \cap [a - t_j, a + U - t_j]. \tag{5}$$

If I_j contains a state satisfying (3), choose the one closest to zero, breaking ties toward the smaller state. Otherwise choose the integer in I_j closest to zero. A protected pixel takes its forced state. Update a after each pixel and u after each row.

Lemma 1 (Exact remaining-row realizability)**.** *For preceding-row states $u_j \in [-B, B]$, the sets in* (4) *are exactly those admitting a legal completion. Each is an integer interval, and $0 \in V_0$.*

Proof. An unprotected successor b permits predecessors in $[b + t_j - U, b + t_j] \cap \mathbb{Z}$. Their union over consecutive $b \in V_j$ gives (4), including $U = 0$. Protected transitions translate intervals. Backward induction gives exactness. The legal source suffix supplies $u_j \in V_j$, including $0 = u_0 \in V_0$. □

Theorem 1 (Linear-time realization of accurate rectangular sums)**.** *For every bounded integer input, protected set, integer $q \geq 1$, and finite $\varepsilon \geq 0$, the backward–forward construction terminates and satisfies* (1)*. It takes $O(HW)$ arithmetic operations and $O(W)$ additional row workspace.*

Proof. Exact feasibility supplies a successor from each $a \in V_{j-1}$. Every transition enforces range and protection. Induction gives $|E_{ij}| \leq B$ and (1). Two scans per row use constant-work interval and residue operations with row-length arrays. □

Workspace excludes input, output, and the full prefix table for verification.

Lemma 2 (Feasible intervals determine fine precision)**.** *Write $I_j = [l, h] \cap \mathbb{Z}$ at an unprotected forward choice. Its feasible pixel interval is $J_j = [s, v] \cap \mathbb{Z}$, where $s = t_j + l - a$ and $v = t_j + h - a$. It contains $\kappa_j = \lfloor v/q \rfloor - \lfloor (s-1)/q \rfloor$ coarse values. Let m count fine pixels in the field, and m_P count protected source values with nonzero remainder. Then*

$$m = m_P + \sum_{(i,j) \notin P} \mathbf{1}\{\kappa_j = 0\} \le m_P + \sum_{(i,j) \notin P} \mathbf{1}\{|I_j| < q\}. \tag{6}$$

Proof. The formula counts multiples of q in J_j, and the forward rule selects one whenever present. Any q consecutive integers contain a multiple. Protected outputs equal their source values. □

Figure 2 maps these intervals in Cairo. Fine pixels appear where the interval excludes every coarse value, with prefix errors bounded throughout.

Encoding the constructed field

Write $Y = qZ + r$, where $Z = \lfloor Y/q \rfloor$ and $0 \le r < q$. Each 16-bit array is compressed with LZMA, preset six, after grouping low bytes then high bytes. Files store stream lengths, missing and protected masks, configuration, geospatial metadata, reconstruction digest, and checksum. Bit-packed masks use zlib; empty masks have no payload.

This separation also yields a finite-field coding bound. Set $N = HW$ and $Q = \lfloor U/q \rfloor + 1$; m is the fine count in (6).

Proposition 2 (Geometry-dependent representation length)**.** *For $q \ge 2$ and known N, U, q, a lossless representation of Y requires at most*

$$\lceil N \log_2 Q \rceil + \lceil \log_2(N+1) \rceil + \left\lceil \log_2 \binom{N}{m} \right\rceil + \lceil m \log_2(q-1) \rceil \tag{7}$$

bits, plus masks and metadata. For $q = 1$, $m = 0$ and the bound is $\lceil N \log_2(U+1) \rceil$.

Proof. Encode quotient, count, fine positions, and remainders by ranks among Q^N, $N+1$, $\binom{N}{m}$, and $(q-1)^m$ possibilities. Each rank uses the ceiling of its set-size logarithm. Decoding forms $Y = qZ + r$; at $q = 1$, only the quotient remains. □

Coarse alphabets reduce quotient cost; fine pixels add positions and remainders.

The band B sets accuracy; q controls storage. For finite nonempty step set $\mathcal{Q}$, let $L_q(X)$ be the complete file length at q. Select

$$q^* = \arg\min_{q \in \mathcal{Q}} L_q(X) \tag{8}$$

All candidates retain Theorem 1's guarantee. Including a reference step prevents rate increase. Ties favor smaller q. Each trial repeats construction and coding.

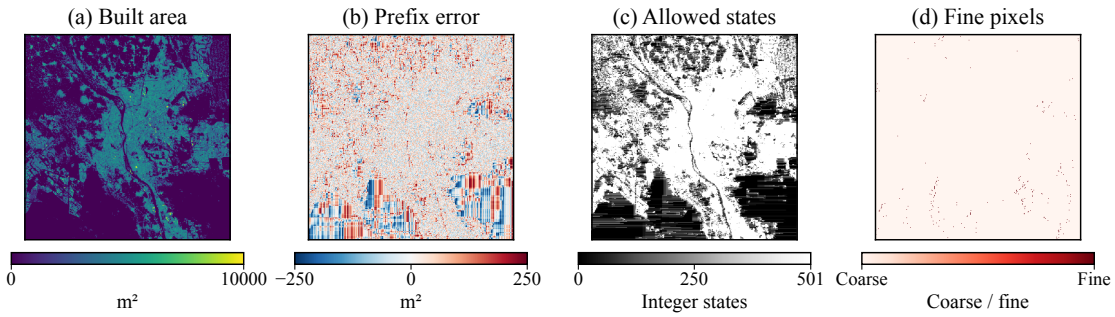


Figure 2: A 512^2 Cairo built-area field with 100 m pixels [19], at budget $\varepsilon = 1000\,\mathrm{m}^2$ and step $q = 200\,\mathrm{m}^2$. (a) Source values. (b) Prefix error bounded by $\pm 250\,\mathrm{m}^2$. (c) Allowed prefix-error states per pixel; darker regions offer fewer choices. (d) Fine pixels retaining nonzero remainders. All panels share the same grid.

Evaluation

Quantities, controls, and complete files

The 2020 Global Human Settlement Layer records integer built area in $[0, 10000]\,\mathrm{m}^2$ per native 100 m pixel [19]. We select the construction rule, fixed step rule, and lossless representation on four development windows. Forty new locations in six continental groups are selected from metadata before pixel access. Each supplies a native 1024^2 window and central 512^2 and 256^2 crops, retaining missing samples. Budgets are 25, 50, 100, 250, 500, 1000, 2500, and $5000\,\mathrm{m}^2$, with fixed step $q = \max(1, \lfloor \varepsilon/5 \rfloor)$.

Water occurrence tests a second quantity [20, 21]. Eight metadata-selected 512^2 windows retain the 1984–2024 product's native 0.00025-degree grid. Values lie in $[0, 100]$; zero is valid and 255 marks missing data. Sum budgets are 5, 10, 25, 50, 100, 250, 500, and 1000 percentage points; q is capped at 100.

Complete files include masks, configuration, geospatial metadata, digest, and checksum. An *eligible file* decodes without the source, preserves the missing mask and protected samples, and meets the integer range and rectangular error budget in (1). We compare six lossless integer representations, lossless JPEG 2000 [22], and SZ3, MGARD, and QPET [4, 11, 14]. Water adds native-byte LZMA and Portable Network Graphics (PNG) files.

For the three scientific compressors, we sweep tolerances 0.49, 1, 4, and 16 with their supported predictor, reordering, or tuning options. Their floating outputs are integer-rounded, clipped to the physical range, and zeroed at missing positions before auditing.

The fixed-margin control changes 2×2 blocks by $(d, -d; -d, d)$, preserving row and column sums, at nine bounds on the integer $|d|$. These controls and RangeRectQuant yield 6600 built-area attempts, 6578 decodes, and 22 QPET encoding aborts. All 392 water files decode. Fresh processes decode files alone. Exact audits of all rectangles cost $O(H^2W)$; prefix bounds also certify eligibility.

A *sum-pyramid control* splits each square's total among four children within their valid ranges. Lossless files store pixels or deviations of child totals from proportional parent shares. We evaluate 2400 complete files on forty 512^2 windows.

Realizing accurate sums within the pixel range

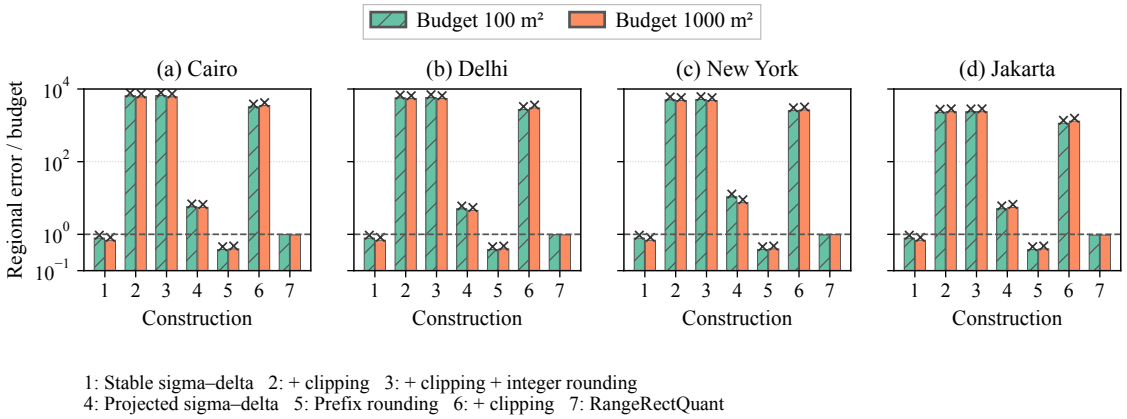


Figure 3: Seven constructions on four cities at two budgets. Bars show the largest rectangle error divided by budget. The dashed line marks the budget; crosses mark integer, range, or regional violations. Only RangeRectQuant meets all requirements in this comparison.

Figure 3 tests realizability. Stable greedy two-dimensional sigma–delta [2] stores rational outputs exactly. Controls clip, clip and integer-round, constrain the alphabet, or round prefixes with optional clipping. All 56 files decode. Stable sigma–delta protects sums with 2.22–6.76% out-of-range pixels. Clipping raises regional error to thousands of times the budget. RangeRectQuant's eight files preserve integer range and every rectangular sum budget.

Rate across locations and budgets

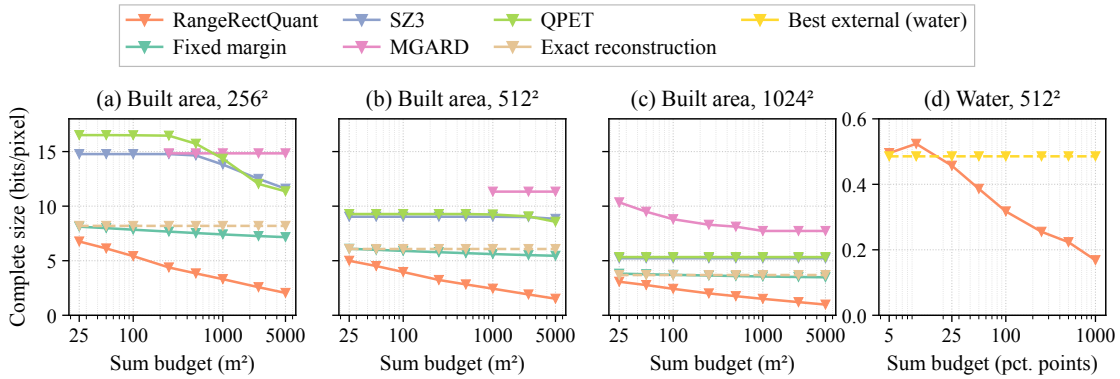


Figure 4: Complete-file rates against lossless, scientific-compressor, and fixed-margin controls. (a–c) Forty built-area locations at three sizes. Each family requires eligibility at every location; gaps mark incomplete coverage. (d) Eight water locations and their best eligible external files. Table 1 adds sum-pyramid controls. Water budgets use percentage-point sums.

All 960 proposed built-area files change pixels and satisfy every requirement. Each is smaller than the best eligible lossless, scientific-compressor, or fixed-margin file,

Table 1: Complete-file rates in bits per pixel, including masked positions. Reduction compares summed bytes with the smallest eligible external files, including 512^2 sum-pyramid controls. Budgets use m^2 for built area and percentage-point sums for water.

Quantity and size	ε	External	RangeRectQuant	Reduction	Wins
Built area, 256^2	100	7.818	5.421	30.65%	40/40
	1000	7.398	3.311	55.24%	40/40
Built area, 512^2	100	5.878	3.965	32.55%	40/40
	1000	4.153	2.438	41.30%	40/40
Built area, 1024^2	100	3.646	2.424	33.50%	40/40
	1000	3.535	1.506	57.39%	40/40
Water, 512^2	100	0.486	0.317	34.66%	7/8
	1000	0.486	0.168	65.35%	8/8

with aggregate reductions of 16.23–72.17% (Figure 4). The reference shifts from exact reconstruction at tight budgets toward fixed-margin blocks as budgets grow.

Adding eligible sum-pyramid files retains forty wins at budgets 100 and 1000, with savings of 32.55% and 41.30% (Table 1). Another 424 scientific-compressor settings on four earlier holdouts leave the best external reference unchanged.

Water savings begin at budget 25 and reach 65.35% at 1000. All eight locations win from 250 onward. Native-byte lossless files are smaller at 5 and 10, defining the precision-and-overhead crossover.

Rate decomposition

For identical fields, quotient/remainder splitting saves 15.20% and 11.03% over direct 16-bit LZMA at budgets 100 and 1000, also beating byte-shuffled coding (Figure 5). Remainders cost 0.070 and 0.086 bits per pixel. Omitting them violates the budget in all 320 fields.

Eight step divisors yield 128 legal files on four development and four earlier holdout windows, with fine choices matching (6) (Figure 5). Coarser steps shrink quotients and increase the fine count. An eight-step probe on the four holdouts uses common metadata and (8). It saves a further 15.04% and 8.30% over the fixed step at budgets 100 and 1000. Search averages 1.20 s per field. Main rate comparisons retain the original fixed step.

An *exact row search* variant minimizes each row's fine-pixel count. With the preceding row's prefix errors fixed, it uses dynamic programming over integer prefix-error states in $[-B, B]$ through the same legal transitions. Ties minimize the sum of squared prefix errors. Chosen prefix errors become the next row's incoming states. This reduces development fine counts by 73.29% and 70.21% at budgets 100 and 1000. Complete files shrink by 0.54% and 0.85%. Quotient storage dominates these files.

Protected samples and scalability

Forty protection configurations pass the joint audit. Nested exact sets cover 0, 0.1, 1, 5, and 10% of valid samples. At budget 100, the largest gives 7.05 bits per pixel and 6.45% fine pixels. Warm construction-plus-audit takes 22–25 ms.

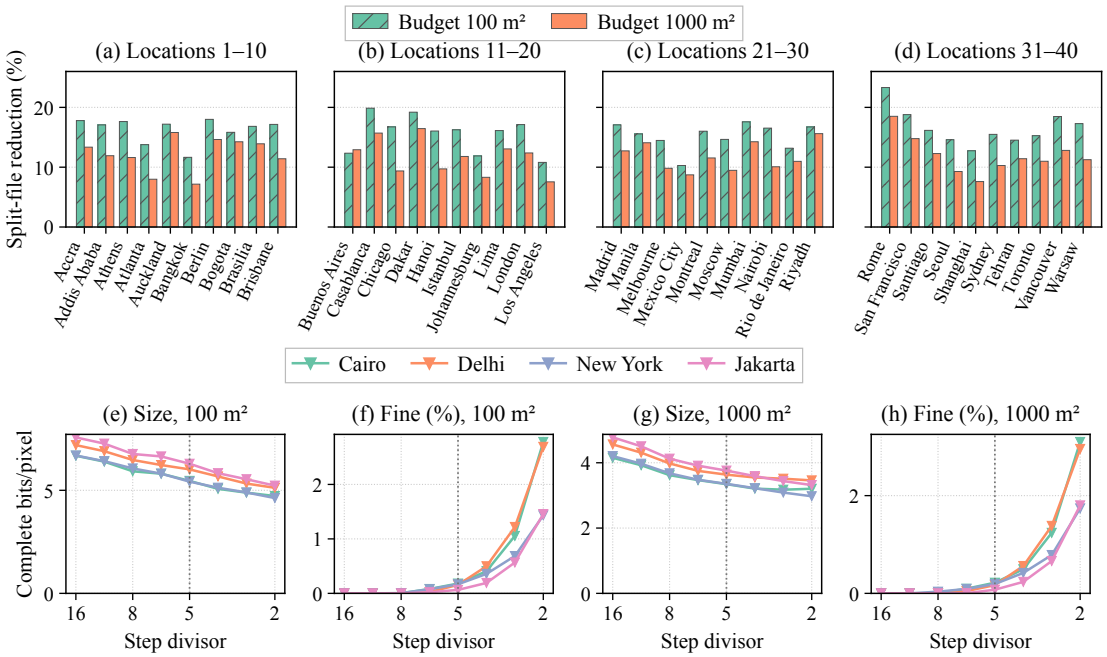


Figure 5: Storage and precision. (a–d) Split-file savings over direct coding of identical fields at all forty 512^2 locations, ordered alphabetically. (e–h) Four earlier holdouts at eight step divisors; $q = \max(1, \lfloor \varepsilon/\text{divisor} \rfloor)$. Dotted lines mark divisor five.

Native London crops reach 4096^2 on an Apple M4 with 16 GB memory. At budget 100, three warmed, single-thread repetitions give median encoding of 7.84 s and decoding of 1.14 s. Construction reaches 18.39 million pixels per second, excluding verification. Exact row search reaches its 180 s time limit at 4096^2 and budget 1000.

Conclusion

One bounded integer field can support every later rectangular measurement under a single absolute budget. RangeRectQuant achieves this by exactly identifying prefix-error states that admit a legal remaining row. The resulting construction takes linear time, preserves designated samples, and enables smaller complete files in the geographic and water-occurrence studies.

Code, data, core experimental results, and extended figures are available in the project repository:
`https://github.com/Marchematics/RangeRectQuant`.

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